

Nathan Perez

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EDUCATION

University of Texas Rio Grande Valley
Bachelor of Science in Computer Science

Edinburg, TX
2021 - 2025

TECHNICAL SKILLS

Languages: C++, SQL, Python, Java, R | Tools: MySQL, Tableau, Excel, Pandas, Figma, Git, Azure

PROJECTS

Phishing URL Detection Analysis (Python, Scikit-learn, Random Forest, XGBoost, pandas, Google Colab)

- Developed and evaluated Random Forest and XGBoost models to detect phishing URLs using a dataset of 235k+ instances
- Performed a comparative analysis using accuracy, precision, recall, and F1-score to assess model performance
- Identified Random Forest as the more robust model, outperforming XGBoost in handling noisy data despite marginal differences in accuracy
- Conducted data preprocessing to clean the data and feature analysis to improve model reliability and model interpretability

Voyage: Travel Itinerary Generator (Figma, HTML/CSS, Python, Flask, PostgreSQL, APIs, GitHub, JavaScript)

- Developed a full-stack web application that generates personalized travel itineraries based on user inputs (budget, location, dates, preferences)
- Integrated multiple APIs including Google Maps, Open-Meteo, Unsplash, and Gemini to enhance data-driven itinerary generation
- Designed and managed backend data handling using the Flask framework and PostgreSQL for database management
- Collaborated using GitHub workflows and designed UI/UX prototypes in Figma

Cloud-Based Media Server (Azure, Cyberduck, Emby, Linux)

- Configured a virtual machine on Azure to function as a cloud-based media server
- Implemented secure file transfers using SFTP (utilizing Cyberduck) and managed server access via SSH
- Deployed and configured Emby media server, including port management, directory binding, and system setup and operations through command-line interface (Linux)
- Applied networking concepts including client-server architecture, access control, and port configuration

SIMULATIONS

Academic Data Analysis Simulations (SQL, MySQL, Tableau, Excel)

- Analyzed structured datasets simulating academic environments, including class performance, course drop rates, and reasons for student attrition
- Wrote SQL to identify trends and patterns in data
- Analyzed relationships between variables to uncover key factors influencing student outcomes.
- Created data visualizations in Tableau to communicate findings and highlight actionable insights
- Performed data cleaning and transformation to prepare datasets for analysis